# DGFC revision: results summary

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## Headline

The DGFC beats both competitors already in the paper, significantly, on every series where a difference is detectable. It is statistically indistinguishable from a modern stochastic-volatility BVAR on 12 of 14 macro series and significantly behind on 2.

- vs the plain BVAR: DGFC significantly better on **14 of 14**
- vs the DLM: DGFC significantly better on **10 of 14**, never worse
- vs the SV BVAR: DGFC significantly worse on **2 of 14**, better on 0

Interval forecasting goes the other way. The DGFC has narrower 95% intervals than the SV BVAR on **11 of 14** series at mean coverage 0.946 against the SV BVAR's 0.964, so it is both sharper and closer to the 0.95 nominal. MSE is a tie at 7 to 7.

Significance throughout is a Newey-West t on the per-vintage CRPS differential, significant at |t| $> 1.96$.

## Macro, Table 2 at saturated k

The full four-metric table is in `macro_table_k14_labeled.csv`. Split here into the two views that matter.

Table 1: CRPS, lower is better.

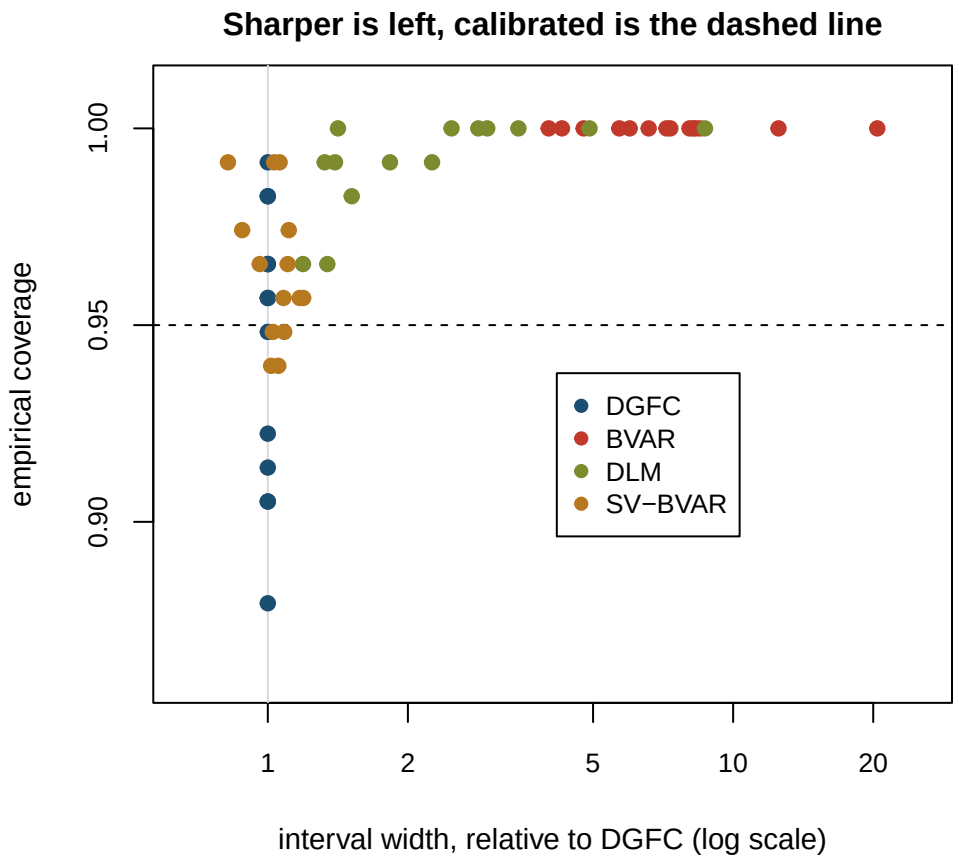
| series | DGFC | BVAR | DLM | SV-BVAR |
|---|---|---|---|---|
| gdpgrowth | 1.261 | 5.653 | 1.389 | 1.214 |
| pcegrowth | 1.299 | 4.693 | 1.265 | 1.220 |
| bfigrowth | 4.426 | 14.566 | 5.304 | 4.569 |
| resinvgrowth | 7.806 | 46.954 | 8.134 | 6.855 |
| ipgrowth | 2.316 | 8.116 | 2.749 | 2.196 |
| capu | 0.568 | 1.684 | 1.693 | 0.545 |
| employ | 0.534 | 1.844 | 0.834 | 0.511 |
| hours | 1.068 | 4.166 | 1.412 | 1.005 |
| ur | 0.125 | 0.327 | 0.552 | 0.111 |
| gdppigrowth | 0.584 | 1.157 | 0.830 | 0.619 |
| pcepigrowth | 0.864 | 1.369 | 0.982 | 0.862 |
| ffr | 0.266 | 0.927 | 0.675 | 0.216 |
| spread | 0.294 | 0.596 | 0.515 | 0.232 |
| realSPgrowth | 14.211 | 100.095 | 13.931 | 13.313 |

Table 2: Coverage against a 0.95 nominal, and 95% interval width.

| series | cov DGFC | cov SV | cov BVAR | cov DLM | sz DGFC | sz SV | sz BVAR | sz DLM |
|---|---|---|---|---|---|---|---|---|
| gdpgrowth | 0.983 | 0.991 | 1 | 0.991 | 11.73 | 12.44 | 97.54 | 16.36 |
| pcegrowth | 0.905 | 0.966 | 1 | 1.000 | 9.50 | 10.48 | 80.75 | 13.44 |
| bfigrowth | 0.922 | 0.940 | 1 | 0.966 | 29.67 | 31.25 | 243.16 | 39.83 |
| resinvgrowth | 0.966 | 0.991 | 1 | 0.991 | 63.60 | 65.62 | 796.07 | 84.27 |
| ipgrowth | 0.966 | 0.948 | 1 | 0.983 | 18.66 | 20.24 | 136.75 | 28.26 |
| capu | 0.957 | 0.948 | 1 | 1.000 | 4.25 | 4.36 | 28.01 | 20.90 |
| employ | 0.957 | 0.957 | 1 | 1.000 | 4.37 | 4.73 | 31.44 | 10.85 |
| hours | 0.991 | 0.974 | 1 | 0.991 | 8.84 | 9.80 | 71.22 | 16.17 |
| ur | 0.948 | 0.966 | 1 | 1.000 | 0.97 | 0.93 | 5.54 | 8.46 |
| gdppigrowth | 0.914 | 0.957 | 1 | 1.000 | 4.13 | 4.84 | 19.72 | 11.71 |
| pcepigrowth | 0.879 | 0.940 | 1 | 0.991 | 5.36 | 5.44 | 21.50 | 12.07 |
| ffr | 0.966 | 0.974 | 1 | 1.000 | 2.66 | 2.34 | 15.91 | 9.18 |
| spread | 0.983 | 0.991 | 1 | 1.000 | 2.47 | 2.03 | 10.60 | 7.32 |
| realSPgrowth | 0.905 | 0.957 | 1 | 0.966 | 83.18 | 99.11 | 1697.70 | 99.00 |

### Sharpness against calibration

Maximize sharpness subject to calibration, so the useful view is both at once. Width is normalized by the DGFC's width for that series, since the fourteen series differ in scale by three orders of magnitude.

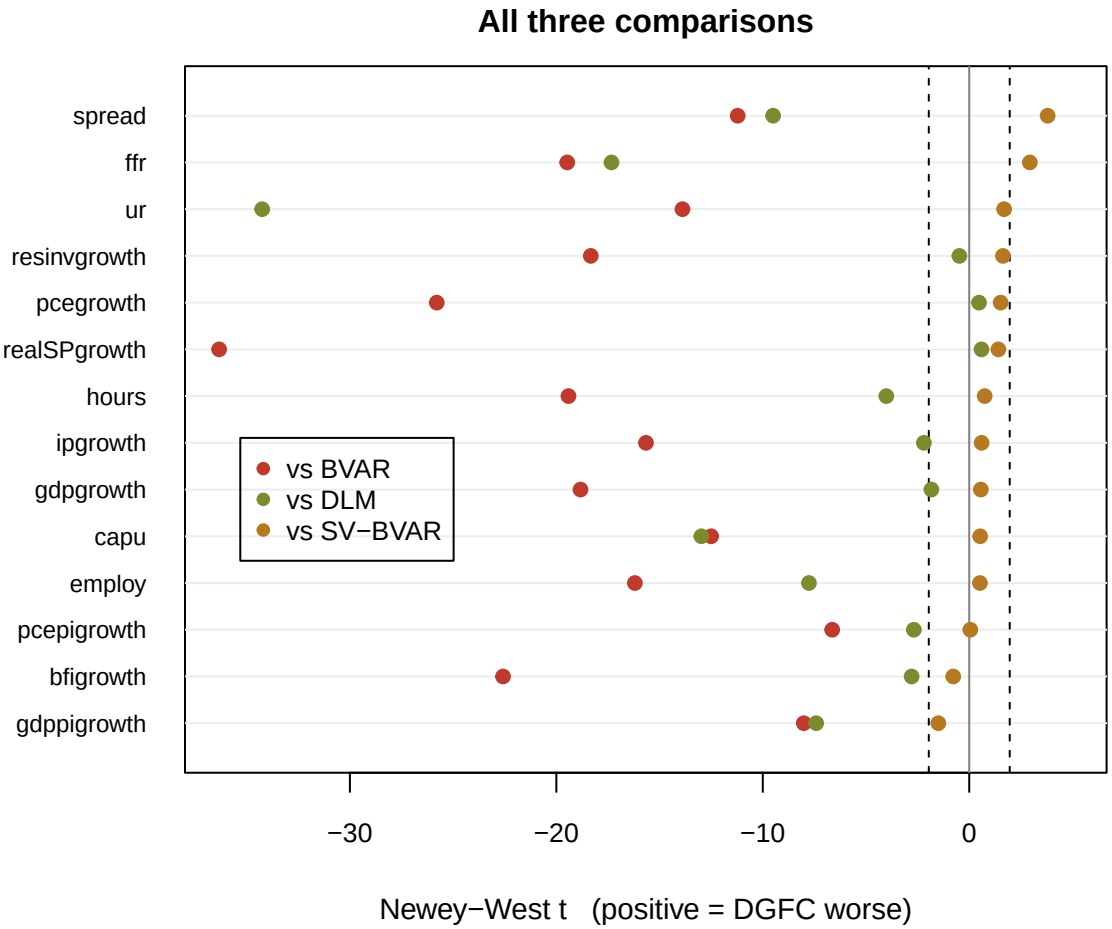



The plain BVAR sits at coverage 1.000 on every series with intervals several times wider, which is uninformative rather than good. The DLM over-covers at a smaller width penalty. The SV BVAR is close to the DGFC on width and slightly over-covers.

## Significance

Table 3: Positive t means the DGFC is worse. Significant at 5% beyond 1.96; Bonferroni across 14 tests is 2.91.

| series | pct_sv | t_sv | t_bvar | t_dlm |
|---|---|---|---|---|
| spread | -21.1 | 3.80 | -11.22 | -9.50 |
| ffr | -18.6 | 2.94 | -19.47 | -17.33 |
| ur | -11.3 | 1.69 | -13.89 | -34.25 |
| resinvgrowth | -12.2 | 1.63 | -18.33 | -0.48 |
| pcegrowth | -6.1 | 1.52 | -25.79 | 0.47 |
| realSPgrowth | -6.3 | 1.41 | -36.33 | 0.59 |
| hours | -5.9 | 0.75 | -19.41 | -4.02 |
| ipgrowth | -5.2 | 0.60 | -15.66 | -2.20 |
| gdpgrowth | -3.7 | 0.56 | -18.83 | -1.83 |
| capu | -4.1 | 0.53 | -12.50 | -12.98 |
| employ | -4.3 | 0.52 | -16.20 | -7.76 |
| pcepigrowth | -0.2 | 0.05 | -6.64 | -2.68 |
| bfigrowth | 3.2 | -0.77 | -22.58 | -2.79 |
| gdppigrowth | 6.0 | -1.50 | -8.01 | -7.41 |



The two comparisons already in the paper produce t-statistics up to 36, so the test has ample power. The nulls against the SV BVAR are genuine, not a power failure.

## Crime, Table 1 at saturated k

DGFC wins CRPS **10 of 10**, interval size **10 of 10**, and MAE **9 of 10**. Mean coverage 0.958 against a 0.95 nominal. The NB-DGLM reaches 0.938 but with intervals 3 to 46 times wider.

Table 4: Crime CRPS by model, with the DGFC's coverage and interval width. Full four-metric table in crime_fcasts_k10_labeled.csv.

| series | DGFC | BVAR | Pois | NB | cov DGFC | size DGFC |
|---|---|---|---|---|---|---|
| Murder | 1.53 | 18.33 | 11.37 | 13.89 | 0.96 | 11.13 |
| Attempted | 1.31 | 38.84 | 3.12 | 4.13 | 0.98 | 9.64 |
| Accessory | 0.30 | 3.56 | 0.66 | 0.67 | 0.99 | 2.61 |
| Manslaughter | 0.49 | 4.66 | 1.13 | 1.18 | 0.99 | 3.55 |
| Non-domestic assault | 114.23 | 13698.82 | 2071.04 | 2392.50 | 0.93 | 769.97 |
| Officer assault | 15.56 | 848.53 | 280.45 | 343.13 | 0.94 | 104.97 |
| Abduction | 3.67 | 102.64 | 17.67 | 19.82 | 0.95 | 27.74 |
| Blackmail | 6.58 | 16.81 | 61.12 | 20.06 | 0.96 | 32.48 |
| Escape custody | 3.21 | 57.19 | 22.87 | 20.31 | 0.94 | 20.80 |
| Resist officer | 36.53 | 2136.75 | 491.60 | 432.57 | 0.94 | 254.24 |

Table 5: Crime, Newey-West t on per-origin CRPS over 284 origins. Negative means the DGFC is better.

| series | t_BVAR | t_Pois | t_NB |
|---|---|---|---|
| Murder | -14.50 | -6.45 | -5.78 |
| Attempted | -20.53 | -6.54 | -4.79 |
| Accessory | -24.01 | -4.70 | -3.47 |
| Manslaughter | -25.18 | -3.20 | -3.63 |
| Non-domestic assault | -84.60 | -12.67 | -11.23 |
| Officer assault | -88.25 | -9.14 | -5.54 |
| Abduction | -21.95 | -10.04 | -6.95 |
| Blackmail | -5.02 | -1.41 | -2.46 |
| Escape custody | -58.54 | -9.05 | -7.43 |
| Resist officer | -52.43 | -7.91 | -9.89 |

DGFC significantly better on 10 of 10 series against the BVAR, 9 against the Poisson DGLM, and 10 against the NB DGLM. Significantly worse on 0.

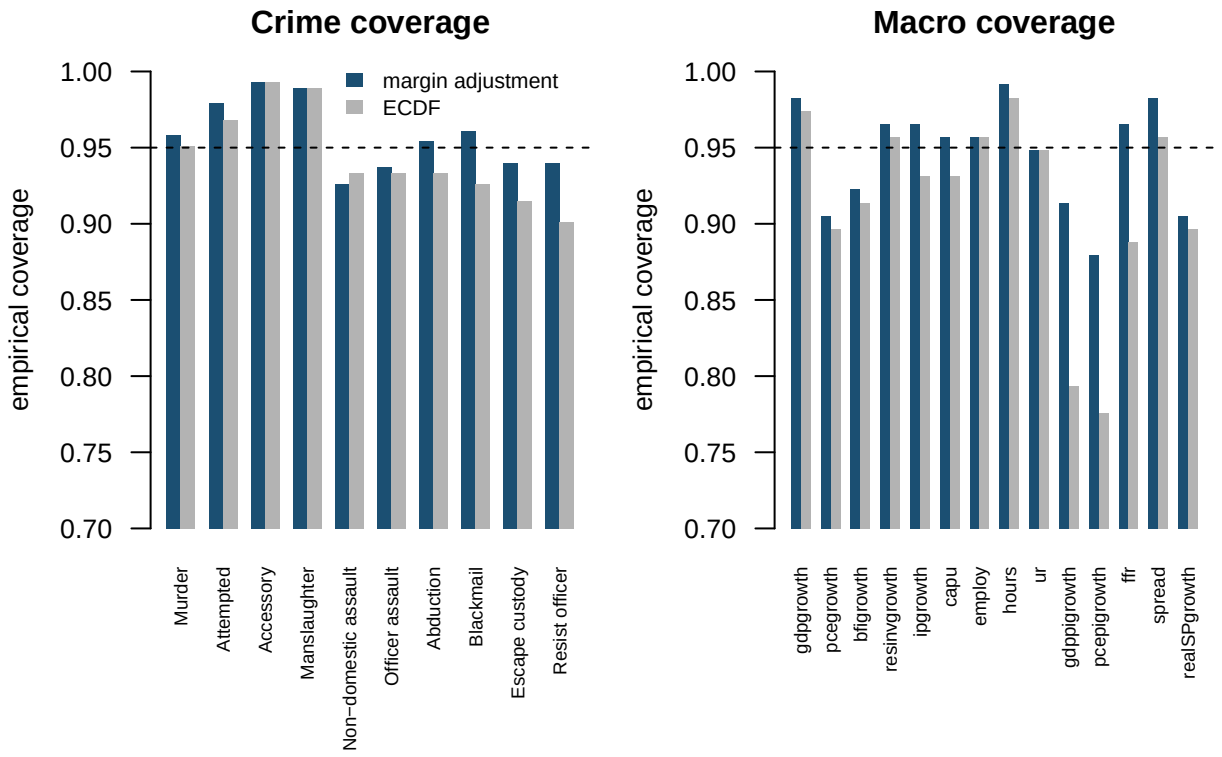
*Per-origin CRPS cached in* `crime_k10_crps_per_origin.rds`*, built 29 Oct 23:52 from* `experiement_crime_stage_8675309/crime_k10_after_loop.RData`*.*

## The ECDF ablation

Freezing the margin at the empirical CDF removes the uncertainty in F while leaving everything else alone. This is the direct answer to what margin estimation buys.

**Crime.** Intervals narrow on 9 of 10 series and CRPS gets worse on 8, by up to 19% on blackmail. Mean coverage falls 0.958 to 0.944.

**Macro.** Intervals narrow on 11 of 14 and mean coverage falls 0.946 to 0.914. CRPS improves on 8 of the 10 growth rates and breaks on capu, ur and ffr at 37%, 99% and 69% worse.

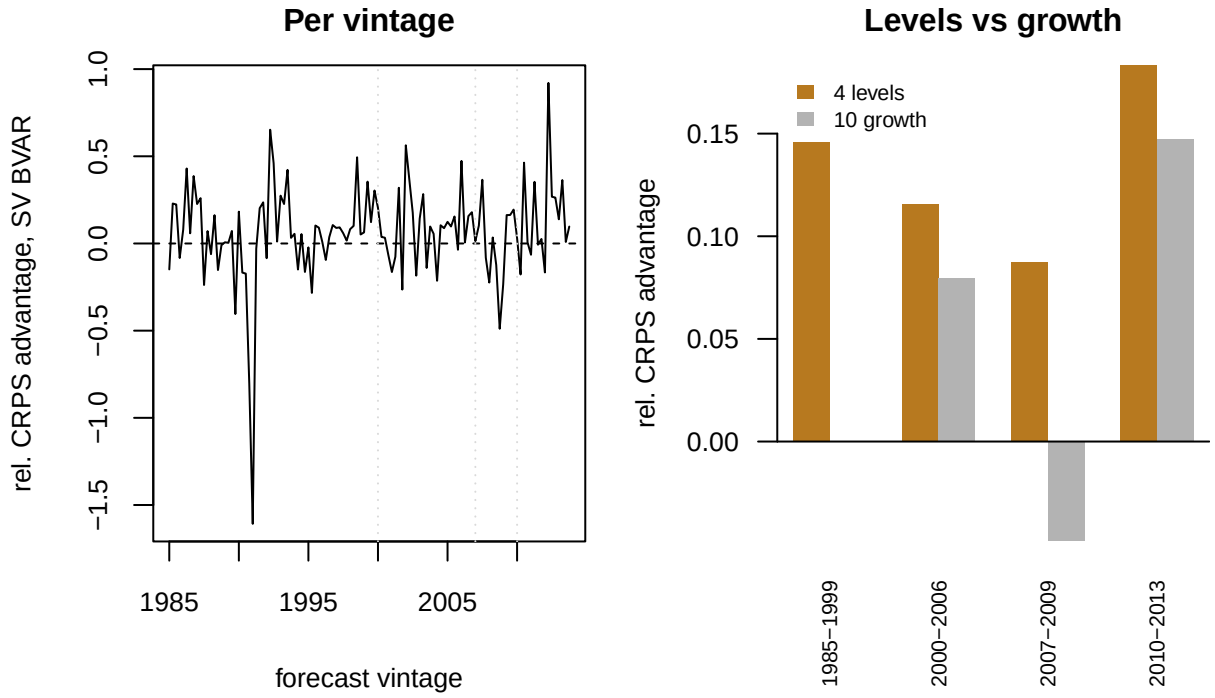



Freezing the margin costs calibration on both exercises. On crime it costs accuracy as well.

## Where the macro loss sits

Table 6: Relative CRPS advantage of the SV BVAR. Positive means the SV BVAR is ahead.

| era | all | levels_4 | growth_10 | vintages |
|---|---|---|---|---|
| 1985-1999 | 0.0419 | 0.1458 | 0.0003 | 60 |
| 2000-2006 | 0.0900 | 0.1159 | 0.0796 | 28 |
| 2007-2009 | -0.0098 | 0.0876 | -0.0487 | 12 |
| 2010-2013 | 0.1578 | 0.1833 | 0.1475 | 16 |



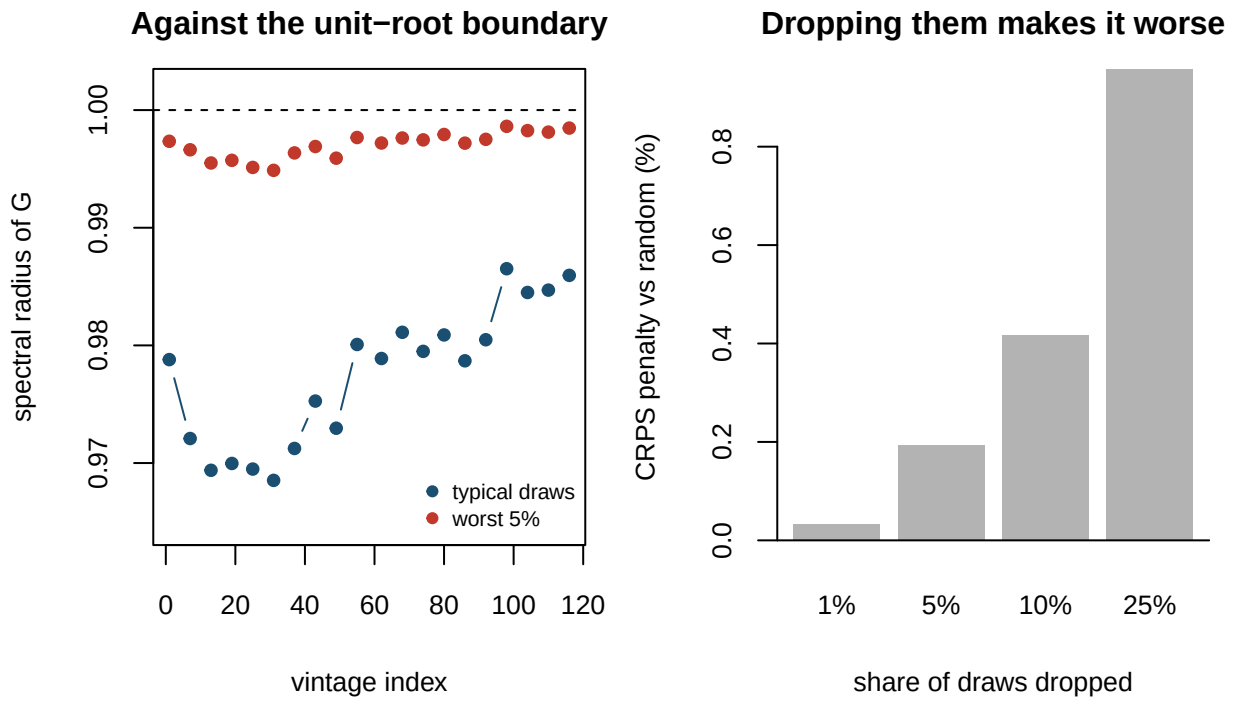

The SV BVAR's advantage is **not** concentrated in the volatile period. It is smallest in 2007 to 2009, the only era where the DGFC leads on growth rates, and the largest single DGFC win in the sample is a 1991 vintage. The split is by series, not by period.

## Mechanism

The latent factor VAR sits at the stationarity boundary across the whole posterior. The mean spectral radius of G is **0.9774** even among the well-behaved draws, rising to **0.9970** in the worst 5%, and the rejection sampler is the only thing keeping it stationary.

The largest latent value is identical in both groups, 4.0198 against 4.0202, so the latent series is not compressed. What differs is the implied scale: 6.6871 against 2.1430, a ratio of 3.12. The stationary covariance of an AR(1) goes as 1/(1 - rhoˆ2), which predicts a ratio of **2.73** from those two spectral radii.

Correlation between boundary proximity and the compressed margin: -0.547.




The draws nearest the boundary are not errors. Removing them degrades CRPS monotonically against a random subsample of identical size, so they are carrying real uncertainty about persistence.

The SV BVAR says the same thing from the other side. `stable = TRUE` discards non-stationary draws and keeps a median of 2828 of 5000, with 47 of 116 vintages keeping fewer than half.

The macro data is not stationary, both models say so. The cost to the DGFC is two series out of fourteen.

## Provenance

| Output | Produced by |
|---|---|
| `crime_fcasts_k10.csv`, `crime_ablation_k10.csv` | crime pipeline, k = 10, 5000/5000 |
| `macro_table_k14.csv`, `macro_ablation_k14.csv` | macro pipeline, k = 14, 5000/5000 |
| `macro_k14_metrics.rds` | macro pipeline post-processing |
| `degenerate_draw_test.rds` | `test_degenerate_draws.R`, 20 vintages |
| MCMC settings (5000 after 5000) | `dgfc_ess_diagnostic.R` |
| DGLM importance-weight degeneracy | `dglm_diagnostic.R` |
| Parallel refactor is faithful | `check_stage1.R` |
| ECDF flag is inert when off | `check_ecdf_flag.R` |

Checked and ruled out: time-varying volatility as the explanation for the SV BVAR's edge; the spline's upper-tail extrapolation as a CRPS cost (`test_pad_mult.R`, median change 0.0%); degenerate margin draws as a CRPS cost (above).

Still open: the latent factor VAR is hardcoded to one lag while every macro competitor uses four. Given that the binding constraint is persistence, that is the remaining candidate, and it is a modelling decision rather than an implementation one.